It can integrate with many databases, data processing elements like Mongodb, Apache Spark, Cassandra, etc, with support for structured and unstructured data and messaging frameworks like Apache Kafka. It enables highly available servers with clustering support, thus reducing downtime and precluding single points of failure using Zookeeper support.

URL: kaaproject.org

SiteWhere: This is an open platform for IoT with good protocol support and powerful device and asset management services, and is licensed under the Common Public Attribution License Version 1.0 (CPAL-1.0). It is designed as a multi-tenant system to host multiple applications in a single instance of deployment. Popular open source solutions like Apache Spark, OpenHAB and InfluDb/Grafana can be integrated by SiteWhere. It can be deployed on public clouds such as Azure, Amazon EC2, service providers based on Ubuntu Juju, Docker or your own server.

URL: sitewhere.org

InfluxData TICK stack: InfluxData provides the TICK stack with four powerful components for time series data management. It is written in the Go language and licensed under MIT terms.

- Telegraf is used for data collection and publishing metrics with the rich set of input, output and service plugins.
- InfluxDb is a time series data base with high availability and high performance without any external dependencies (unlike other open source time series models like OpenTSDB based on Hbase or KairosDB based on Cassandra). It has good support for down sampling high precision data and data retention policies.
- Chronograf is a visualisation tool for exploring data.
- Kapacitor is used for data processing with alert management and ETL jobs.

URL: influxdata.com

Mosquitto: This is an Eclipse IoT project implementing a MQTT broker with v3.1.1 and v3.1 support for backward compatibility. It is licensed under the Eclipse Public License - v1.0 and EDL 1.0. It also provides client libraries and powerful reference clients. Popular cloud services like CloudMQTT are built on top of Mosquitto.

URL: mosquitto.org

CloudFoundry: This open source PaaS platform is a Linux Foundation collaborative project and has been released under the Apache 2.0 License. It is available as a service from a few commercial providers like Bluemix or as an open source product to be deployed on one's own servers. It supports the full life cycle of application development in various languages for IoT needs, and comes with good service support like data storage, messaging, application development and APIs for mobile apps.

URL: cloudfoundry.org

Connectivity

Paho: This is an Eclipse IoT project offering MQTT clients, licensed under EPL v1.0 and EDL v1.0. It offers synchronous and asynchronous API mode in various languages like C, C++, Java, Python, Go, JavaScript, and .Net (C#). It also provides an Android service for building mobile apps and reference apps, which can be customised according to an application's needs.

URL: eclipse.org/paho

Eclipse Californium: This is divided into a few sub-projects. The core comes with the Java API model for the CoAP client and server design, and a few reference apps. It can be built as a Maven project and can be embedded in other Java applications. The Scandium sub-project implements DTLS 1.2 for secure CoAP. It also provides a proxy library and reference app for CoAP-HTTP bridging. These are dual licensed under EPL v1.0 and EDL v1.0.

URL: eclipse.org/californium

Bluez and bindings: This is an official Linux Bluetooth protocol stack and comes with a user space library and tools. It supports various classical protocols like L2CAP, RFCOMM, SDP, etc, as well as LE protocols, profiles like GATT, and GAP. It has Python bindings via PyBluez and Node.js packages. Bleno and noble via npm are available for BLE advertising in peripheral mode and discovery in central mode. These node packages are built on top of Bluez.

URL: bluez.org, karulis.github.io/pybluez/, github.com/sandeepmistry

CETIC 6LBR and Linux WPAN: 6LowPAN/RPL Border Router (6LBR) can interconnect end devices running on 6LowPAN with the Internet, by bridging the 802.15.4 network with IPv6 on the gateway side. It supports various 15.4-capable targets like OpenMote, TI CC25xx/26xx platforms and Linux hosts like Raspberry Pi, and can run in Bridge mode or Router mode. It is based on the Contiki project and licensed under similar terms.

Linux WPAN enables the 802.15.4 stack at the kernel level and user space tools for 6LowPAN based development on Linux.

URL: http://cetic.github.io/6lbr/,github.com/linux-wpan

Wireshark filters: Wireshark, a well-known network protocol analyser and packet generator, is licensed under the GNU GPL. It has rich filter supports for Internet protocols like MQTT, CoAP, HTTP, Websockets, etc. It can also analyse Bluetooth (classic, LE) packets and IEEE 802.15.4 traffic. END

References

Official sites, GitHub repos, and wiki pages of mentioned projects

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